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An optical device for recording glottal adduction-abduction during normal speech

The growing interest in the motor theory of speech perception [3] has resulted in a renaissance of articulatory phonetics. The earlier study of the physiological parameters of phonemes has been resumed with improved electronic instruments, and several new areas have been opened. One of the particular preoccupations of the moment is the glottis, and a number of devices are under development to permit recording the movements of the vocal folds. The vibrations have of course long been inferred from their acoustic output, orthophonic and pneumatic devices having made their appearance in the last century. These have since been replaced by various kinds of electronic oscillographs, using air or contact microphones, and more recently by the sound spectrograph. There are, however, certain glottal movements of phonemic significance that escape detection by these methods. These include the gross movements that result in the adduction and abduction of the vocal folds and, in the vibratory mode, the "edge vibrations" recently discussed by LISKEB and ABRAMSON [4]. What is needed now is obviously a device capable of tracking glottal action independently of acoustic output.

Three purely electronic schemes have been tried or considered, but each has its drawbacks. One was developed by F. FABRE [2] and involves the transmission of a radio signal through the larynx. Oscillograms are obtained of modulations of this carrier caused by the varying capacitance of the glottis as a function of the proximity of the folds. The weakness of this device is that it cannot be applied to the study of normal speech. The larynx as a whole moves about considerably during articulation, thereby precluding the possibility of keeping the signal centered through the pertinent area. Another difficulty with FABRE's circuit as described is that it responds only to changes and thus will not indicate the position of the folds when they are at rest or moving very slowly. Another possibility would be to use ultrasonics, but here again directing the beam through the folds is essential for obtaining meaningful recordings. Electromyography is also available, but the complexity of the muscular activity involved in the vibration and other movements of the vocal cords makes it virtually impossible to equate any single parameter with occlusion or opening, let alone phonation, and we hardly need mention the insertion of appropriate electrodes!

The most promising systems appear to be optical, and these divide themselves into two classes, the first being motion picture photography and the second the measurement of the amount of light projected by an appropriate lamp either

upward or downward through the larynx and valved by the opening and closing glottis. High speed motion picture photography has apparently solved the problem of the "edge vibrations" — see LISKEB [4], p. 147 — and stroboscopy provides the means of "stopping" the vibrating folds at any point in their cycle or slowing the movement down to an analyzable rate of speed. However, all laryngoscopic methods developed so far preclude the observation of the larynx during normal speech, as viewing must be effectuated through the open mouth.

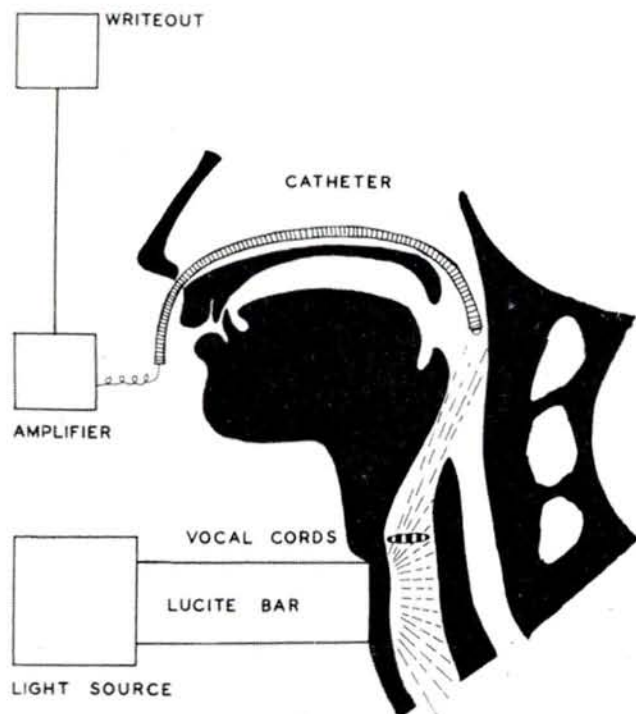


Fig. 1. Glotlux (Block Diagram)

The forthcoming Cooper system, as announced in the LISKEB and ABRAMSON article [4] (p. 418, note no. 52) is expected to overcome this difficulty. The second class of optical systems is exemplified by SONASSEN's glottal transilluminator [7], in which a light is projected transcutaneously into the trachea at the fossa jugularis, some of which diffuses upward through the larynx to impinge on the end of a light-conducting rod leading from the mesopharynx out through the mouth to a photomultiplier tube. This method is evidently very effective in recording the vibrations of the folds during the phonation of a musical note while the mouth is open. However, the same drawback continues to exist for normal speech as in the case of the laryngoscopic methods. F. S. COOPER has since developed a transilluminator—see LISKEB [4], p. 417 — that is the reverse of SONASSEN's device: Light is conducted through a rod and projected downward from the mesopharynx through the larynx, and a phototube is pressed against the fossa jugularis to

measure what is diffused through the skin in that area. From the point of view of normal speech, this would not seem to represent any improvement, as it interferes with buccal articulation.

The apparatus developed by the present writers bears the name "Glotlux" and was inspired by SONASSEN's. Our experience with catheters inserted through the nose for recording intraoral air pressures during normal speech [6] suggested mounting a miniature photocell in the end of such a catheter to measure the

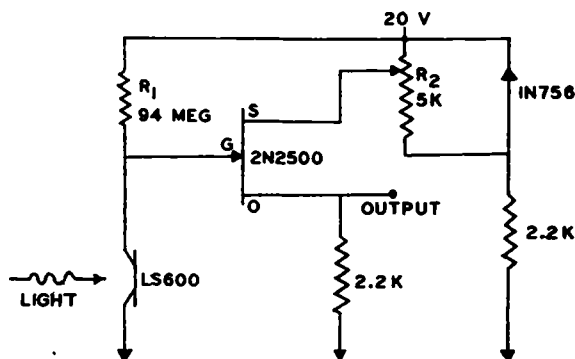


Fig. 2. Photoelectric amplifier circuit

amount of light being valved into the mesopharynx by the vocal folds and emanating from a light source at the fossa jugularis. A block diagram is shown in Fig. 1. The light comes from a 500-watt projector with a bar 2'' in diameter to conduct it to the subject's neck. With this intensity, it is not necessary to work in a darkened room. When the vocal folds are wide open, as during deep inhalation, the glow is clearly visible in the back of the mouth. To eliminate any ripple due to fluctuations of the projector's normal A. C. light source, the light has been converted to D. C. by using an appropriate filtered power supply. The photocell is Texas Instruments LS-600 mounted in the end of a Lee DC-10 catheter. The circuit for the amplifier is shown in Fig. 2. Writeout is on a Honeywell "Visi-corder" (906-129 XFGH) with a direct connection from the amplifier to a recording galvanometer (Honeywell M 400-350). The sensitivity of the system is predicated on the fact that our own present interest is in recording the adduction and abduction of the vocal folds rather than their vibrations — the circuit can be modified appropriately as our interests change.

This device lends itself to a number of interesting investigations. High in priority in our present plans is a study of vocal fold adduction and abduction during the interval corresponding to buccal occlusion or constriction of phonemically voiced consonants that have become partially or completely voiceless due to phonetic context, e. g., the /b/ of *absent*. This is but one aspect of the quest for the physiological invariants of the so-called lenis/fortis distinction: It is now well-established that the "lenis" consonants [b], [d], [g], [v], [z], and the like, have lower buccal air pressures than their "fortis" cognates [p], [t], [k], [f], [s] etc., and this is thought to be a function of glottal valving [5]. Assuming pulmonary thrust to be more or less constant for all consonants, the adduction of the vocal

folds, with or without phonation, would pass less air per unit time into the supra glottal cavities and the total build-up would thus be less for /b/ than for /p/, etc. That the folds are indeed adducted for the "lenis" articulations, even in the absence of phonation, has yet to be positively established, but preliminary recordings indicate that this hypothesis is correct. Furthermore, the folds are adducted for the sonants even in whispered speech, at least in the samples obtained from the one subject to make recordings so far. Another question to be explored is the occurrence of the glottal stop as a supporting articulation to reinforce the buccal occlusion of final stops in arresting the pulmonary beat stroke associated with an utterance. We also intend to investigate the occurrence of glottal adduction as a syllable boundary marker in cases of contiguous vowels. In short, there are a number of studies to be undertaken with this device quite apart from glottal vibration.

The successful utilization of Glotlux will depend on how well we are able to reduce a number of variables affecting the fidelity of the recordings. These are divided into two groups, the first affecting the amount of light valved by the vocal folds and impinging upon the photoelectric cell, and the second affecting the sensitivity of the system. Aside from glottal aperture, the principle factor determining the transmission of light through the larynx is the position of the larynx itself. As we observed earlier, it moves about considerably during normal speech. It is raised, depressed, and tilted, and in speech these movements are principally functions of vowel color and musical pitch. Transmission is also affected by variations in thickness of the folds, which in turn is a function of tension. Phonetically, both pitch and intensity are related to this. The light can also be partially absorbed or interrupted by the false vocal cords. It has recently been suggested that movements of the latter are syncretic to voluntary laryngeal activity [1]. Finally, movements of the root of the tongue and of the pharyngeal walls, both of which are functions of phonetic context, as well as of deglutination, will affect the amount of light arriving in the mesopharynx. The solutions to these problems are quite simple. We will have to utilize highly controlled samples of speech, ideally containing but a single open or neutral vowel, uttered at a constant pitch (with the subject following a tone monitored through a headset) and at a constant level of intensity. In the initial phases of study, consonants involving lingual articulations will be avoided, which will temporarily restrict us to bilabials and labiodentals.

The factors affecting the performance of the photoelectric system are as follows: Maintaining a fixed position in the mesopharynx is essential, as all photoelectric cells are directional to some extent at least. By far the most pronounced displacements noted in trial runs have been due to movements of the portion of the catheter and leads outside from the nose to the amplifier. We have succeeded in eliminating these by immobilizing the subject's head in relation to the amplifier and by taping the catheter to his upper lip. Slight displacements are also caused by movements of the upper pharyngeal walls, and these appear to be caused by the contraction of the glossopalatal and pharyngopalatal muscles, that effectuate velar occlusion. Displacement here can be minimized by restricting utterances to

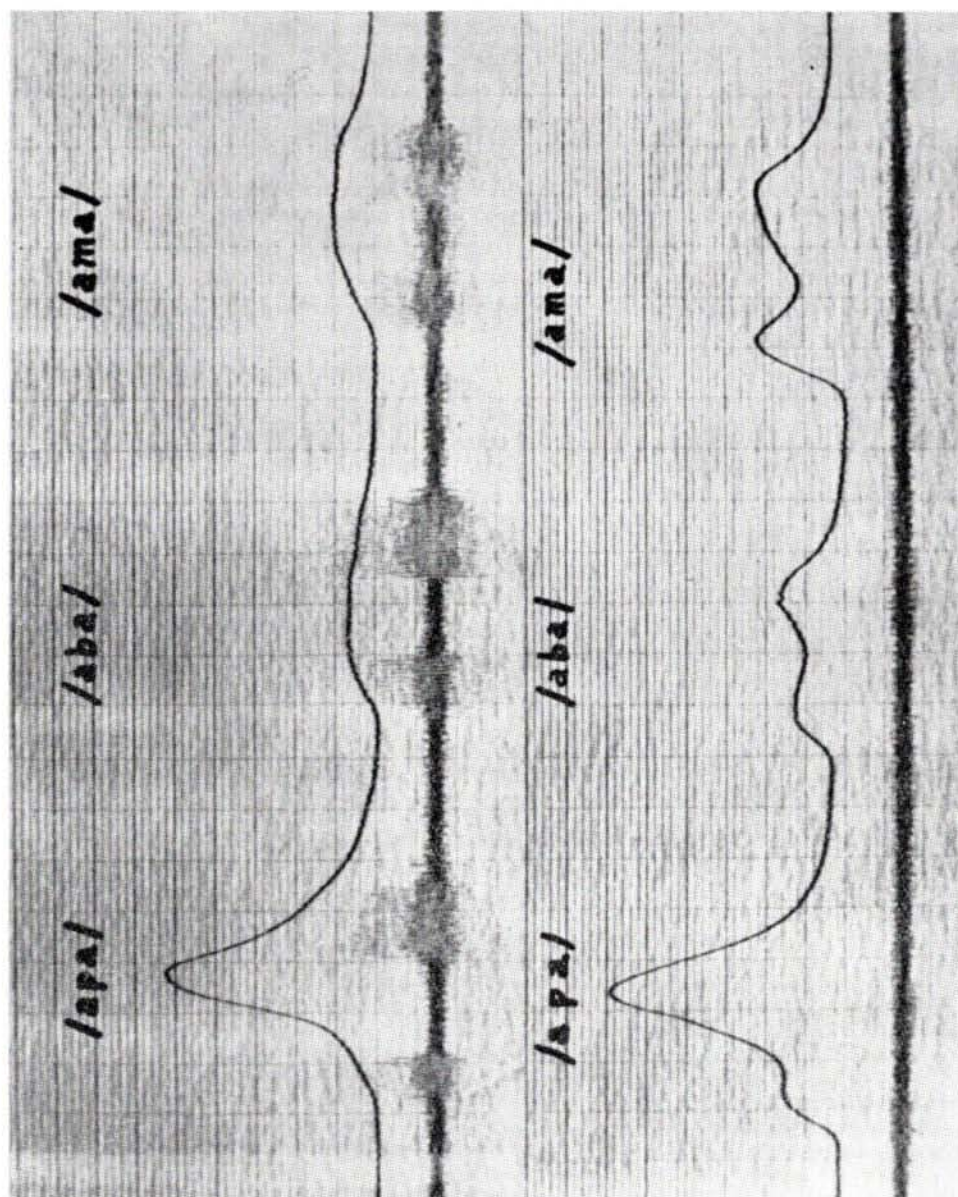


Fig. 3. Recordings of spoken and whispered labial consonants

purely oral vowels in association with oral consonants. This will of course be more easily accomplished in a language such as French, in which oral vowels are articulated with the velum closed, than in American-English, in which all vowels are uttered with at least some degree of nasality. The sensitivity of the cell may

also be affected by accumulations of mucus on the lens. Periodic inspection and cleaning in position with a long cotton-tipped swab dipped in alcohol is necessary to keep it free. Finally, the photoelectric cell is somewhat sensitive to variations in heat caused by inhaling air at lower than body temperatures. This difficulty has been eliminated by providing the subject with an air supply heated to body temperature from an electric heater and blower connected to a Bennett face mask.

Figure 3 shows two sample recordings of /apá/, /abá/, and /amá/, the first one spoken and the second whispered. Two traces are shown in each: The upper one is from Glotlux, and the lower one represents the acoustic output of the glottis as recorded from a button microphone taped to the subject's neck over one of the quadrilateral plates of the thyroid cartilage. The latter provides reference points for the interpretation of the upper trace. It is interesting to note that the upper traces of the two recordings are essentially the same in spite of the fact that the /b/ and /m/ of the first are fully voiced, while in the second they are not. The vocal folds thus appear to adduct for these consonants, even in the absence of phonation. The vibrating folds, which valve a relatively small amount of light into the upper cavities, account for the slight rise in the Glotlux trace during the vowels and voiced consonants. The higher amplitude of whispered /b/ and /m/, as compared to their respective voiced versions, is explained by the fact that although the vocal folds are clearly adducted in both cases, a small aperture persists throughout the former, whereas phonation entails periodic occlusion, and this results in the valving of differing amounts of light.

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